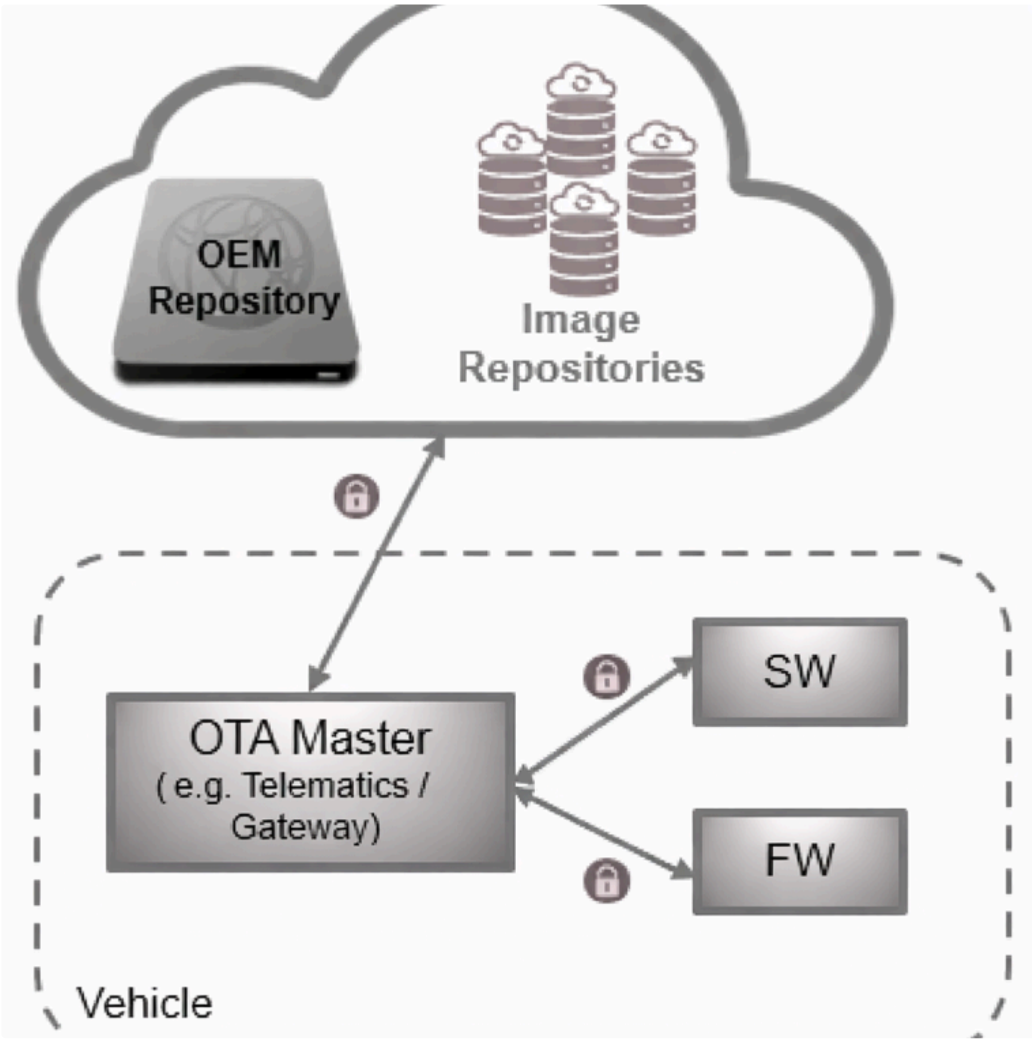
[OTA] Defining Scope, Brainstorming

Introduction:

We need to build a generic mechanism to update ECU/HPC SWs and FWs during runtime. Keeping in mind that the current system is executed and fully available from the following perspectives:

- Already deployed versions should be fully functional unless the newly installed software is activated
- The software should be downloaded and installed in background.
- The system should also ensure that the interruptions or corruptions while installation should continue in multiple driving cycles.
- The system should ensure, verify the authenticity and integrity of the new software.
    - Once, the verification process is completed, the respective hardware should be able to activate the new update.
    - This activation can have some (minimal) downtime and can require special ECU mode, therefore the activation should be process in a safe-state possibly, can be short parking, in parking (user) state, etc.
- If the activation gets failed, it should roll back to the previous version of the software, this means the hardware should ensure that the previous version is still available even the new software is activated.



Scope:

The scope of this work is to design and implement a Over-The-Air (OTA) update mechanism that enables secure, reliable, and resumable updates of SW apps and ECU FW during normal vehicle operation. The mechanism shall allow updates to be downloaded and prepared in the background while the currently deployed software remains fully functional, and shall support controlled activation and rollback to ensure system safety and availability.

1. Applicability and Update Coverage:

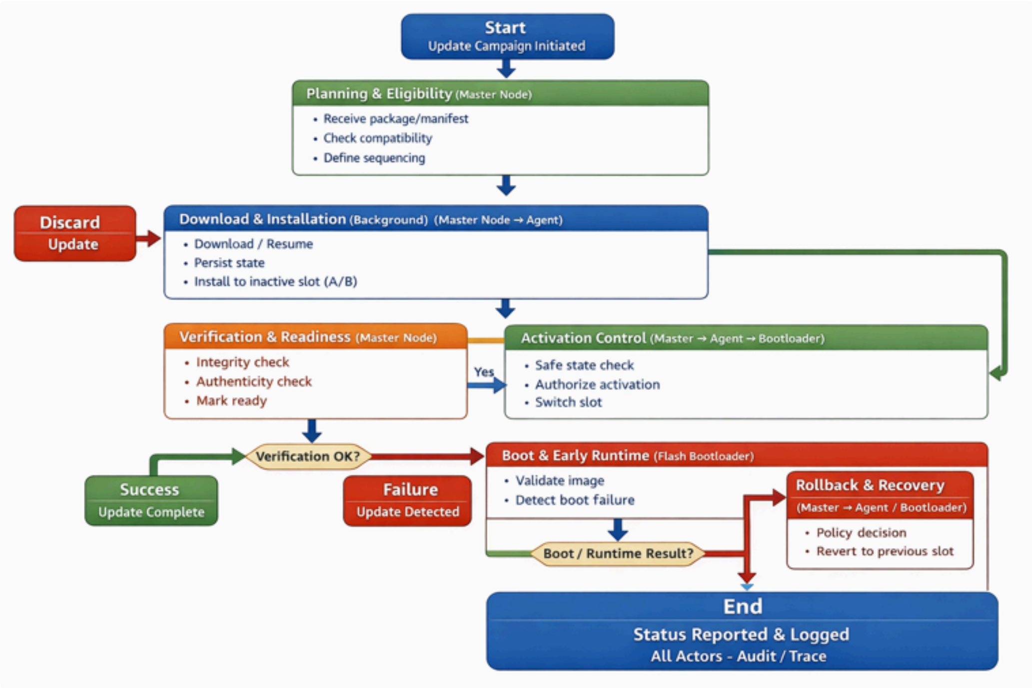
The OTA mechanism applies to the following update targets:

- Software applications and services running on OS-based ECUs and the HPC
- ECU firmware, including single-binary systems such as FreeRTOS or bare-metal ECUs, including safety-related firmware

The OTA mechanism does not apply to HPC firmware updates, which are handled through manual or service procedures outside the scope of this mechanism.

2. Actors in Scope

The OTA update mechanism is structured around three primary actors, each operating at a distinct system layer and lifecycle phase:



a. Master Node

The Master Node acts as the system-level OTA orchestrator. It coordinates update campaigns across the vehicle, evaluates update eligibility using manifest information, and governs the progression of updates through their lifecycle, including authorization of activation based on system and vehicle conditions.

b. Agent

The Agent is a node-local OTA executor deployed on ECUs and compute nodes capable of runtime software execution. It performs background download, installation, verification, and local activation preparation while ensuring that currently active software remains unaffected until activation is explicitly triggered.

c. Flash Bootloader (Hardware specific)

The Flash Bootloader is the boot-time activation and recovery authority, primarily for ECUs running single-binary firmware. It is responsible for firmware activation, boot-time verification, image selection, and rollback to a previously known-good version in case of activation or boot failure.

3. Runtime Update Behavior (A/B slotting):

The OTA mechanism shall support background download and installation of updates while the vehicle system is fully operational. Already deployed and active software shall continue to execute normally until the new software is explicitly activated. Installation shall occur in a non-active or staging area to prevent interference with the running system.

The mechanism shall tolerate interruptions such as power loss, vehicle shutdown, or network loss. Update progress and state shall be persisted such that installation and preparation can safely resume across multiple driving cycles without risking system corruption.

4. Authenticity, Integrity, and Trust

All updates shall undergo mandatory authenticity and integrity verification prior to activation. Only software and firmware verified to originate from a trusted source and confirmed to be unmodified shall be eligible for activation. Verification shall be enforced as a hard gate, ensuring that unverified or corrupted updates are never executed.

5. Activation and Safe-State Control

Activation of new software or firmware may require temporary downtime, special ECU modes, or system restarts. The OTA mechanism shall therefore separate installation from activation and ensure that activation is only performed when the system is in an approved safe state, such as parking or other permitted vehicle states, and when sufficient power and stability are available.

6. Rollback and Recovery

The OTA mechanism shall guarantee the availability of a previously deployed and known-good version of the software or firmware during and after activation. In the event of activation failure or post-activation validation failure, the system shall automatically revert to the previous version to restore normal operation. Rollback shall be deterministic and shall not require manual intervention.

7. Update Lifecycle Management and Visibility

The OTA mechanism shall manage the full update lifecycle, including download, verification, staging, activation, validation, completion, and rollback. It shall provide persistent status reporting and diagnostic visibility at both the node and system levels to support monitoring, auditing, and post-event analysis.

RACI Matrix for the actors

Phase	Activity	Master Node	Agent	Flash Bootloader

Planning & Eligibility	Receive update package and manifest	Owns	No role	No role
	Interpret compatibility information	Owns	No role	No role
	Determine eligible ECUs / nodes	Owns	No role	No role
	Define sequencing and timing	Owns	No role	No role
Download & Installation (Runtime)	Download update in background	Owns	No role	No role
	Resume download after reboot / power loss	Owns	No role	No role
	Persist update state across driving cycles	Owns	No role	No role
	Install into inactive / staging area	No role	Owns	Consulted
Verification & Readiness	Verify integrity (hash/checksum)	Owns	No role	No role
	Verify authenticity (signature/trust)	Owns	No role	No role
	Mark update as ready for activation	Owns	No role	No role

	Enforce “no activation without verification”	Owns	No role	No role
Activation Control	Determine safe activation window	Owns	Consulted	No role
	Authorize activation	Owns	Informed	Informed
	Trigger activation locally	No role	Owns	Consulted
	Switch active image / slot (firmware ECUs)	No role	Consulted	Owns
	Enter ECU update / special mode	No role	Owns	Consulted
Boot & Early Runtime	Validate image at boot	No role	No role	Owns
	Detect failed boot	No role	No role	Owns
	Select fallback image	No role	No role	Owns
Post-Activation Validation	Run health and validation checks	No role	Owns	No role
	Report success or failure	No role	Owns	No role
	Confirm update completion	Owns	Consulted	No role
Rollback & Recovery	Detect runtime failure	No role	Owns	No role

	Detect early boot failure	No role	No role	Owns
	Decide rollback policy	Owns	Consulted	No role
	Execute rollback (runtime)	No role	Owns	Consulted
	Execute rollback (boot-time)	No role	No role	Owns
Status & Audit	Aggregate campaign status	Owns	Consulted	No role
	Report per-node progress	No role	Owns	No role
	Maintain audit and trace logs	Owns	Owns	No role